Numerical soln of wave eqn [Hyperbolic eqn]

The standard equation is of the form

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \longrightarrow (1)$$

phase speed $c^2 = \frac{T}{\rho}$ Tension. Ratio of tension to mass/unit length of string. ρ: density of the string.

Subject to initial conditions: $u = f(x)$, $\frac{\partial u}{\partial t} = g(x)$

$0 \le x \le l$ at $t = 0 \Rightarrow u(x,0) = f(x)$ & $u_t(x,0) = g(x)$

& the boundary conditions

$u(0,t) = \phi(t)$, $u(l,t) = \psi(t)$.

Consider a rectangular mesh in the x-t plane spacing h along x direction & k along time t direction. Denoting a mesh point $(x,t) = i,j$ as simply.

Substitute $\frac{\partial^2 u}{\partial x^2} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2}$ &

$\frac{\partial^2 u}{\partial t^2} = \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2}$ using

finite difference approximation in the standard eqn (1). After simplification it reduces to $u_{i,j+1} = 2(1-\alpha^2c^2)u_{i,j} + \alpha^2c^2(u_{i-1,j} + u_{i+1,j}) - u_{i,j-1} \longrightarrow (2)$ where $\alpha = \frac{k}{h}$.

The coefficient of $u_{i,j}$ vanishes if $\alpha c = 1$ then (2) $\Rightarrow$

$$u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1} \longrightarrow (3)$$

Now by central difference approximation

$$\frac{\partial u}{\partial t} = \frac{u_{i,j+1} - u_{i,j-1}}{2k} = g(x) \text{ (Given)}$$

$$\Rightarrow u_{i,j+1} = u_{i,j-1} + 2k g(x) \quad \text{at } t=0.$$

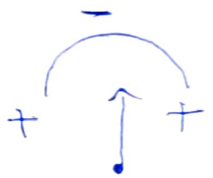
$$\Rightarrow u_{i,1} = u_{i-1} + 2k g(x) \quad \text{for } j=0. \longrightarrow (4)$$

~~This gives the values of~~

When $g(x) = 0$ $\quad u_{i,1} = u_{i,-1}$

then (3) $\Rightarrow$ (For $j=0$). $u_{i,1} = \frac{1}{2}[u_{i-1,j} + u_{i+1,j}]$

~~$u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1}$~~



Note: (1) For $\alpha = \frac{1}{c}$ the soln of (2) is stable & coincides with the soln of standard eqn

(2) For $\alpha < \frac{1}{c}$ the soln is stable but inaccurate

(3) For $\alpha > \frac{1}{c}$ the soln is unstable.

(4) The eqn (2) converges for $\alpha \leq 1$ i.e $k \leq h$

(1) Evaluate the pivotal values of the elliptic eqn $u_{tt} = 16 u_{xx}$ taking $\Delta x = 1$ upto $t = 1.25$. The boundary conditions are $u(0,t) = u(5,t) = 0$, $u_t(x,0) = 0$ & $u(x,0) = x^2(5-x)$.

Soln Compare $u_{tt} = 16 u_{xx}$ with $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

$c^2 = 16 \Rightarrow c = 4$

Given $h = 1$

Substitute the derivatives by finite difference approximation, the given eqn reduces to

$$u_{i,j+1} = 2(1-16\alpha^2) u_{i,j} + 16\alpha^2 (u_{i-1,j} + u_{i+1,j}) - u_{i,j-1} \longrightarrow (1)$$

where $\alpha = \frac{k}{h}$ [& $\alpha = \frac{1}{c}$]

$$\Rightarrow \frac{1}{c} = \frac{k}{h} \Rightarrow \frac{1}{4} = \frac{k}{1} \Rightarrow \boxed{k = \frac{1}{4}}$$

upto 1.25

Given $u(0,t) = u(5,t) = 0 \Rightarrow$ when $x=0$ & $x=5$ the values of u are zero. i.e I & last column are zero.

For I row:

$u(x,0) = x^2(5-x)$

When $x=1$ $\quad u(1,0) = 1^2(5-1) = 4$

$x=2$ $\quad u(2,0) = 2^2(5-2) = 12$

$x=3$ $\quad u(3,0) = 3^2(5-3) = 18$

$x=4$ $\quad u(4,0) = 4^2(5-4) = 16$

For II row

$u_t(x,0) = 0 \Rightarrow u_{i,1} = \frac{1}{2}[u_{i-1,0} + u_{i+1,0}]$

From this we can find $u_{11}, u_{21}, u_{31}, u_{41}$

Solve this eqn numerically for one half period of vibration, taking $h=1$ & $K=\frac{1}{2}$.

Soln Here $c^2=4 \Rightarrow c=2$

$\therefore$ The difference eqn for the given eqn. after simplification is

$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1}$, which gives a convergent soln $\alpha \leq 1$ i.e $K \leq h$

Since $u(0,t) = u(4,t) = 0 \Rightarrow$ I & last Columns are zero

Since $u(x,0) = x(4-x)$

$x=1 \quad u(1,0) = 3$

$x=2 \quad u(2,0) = 4$

$x=3 \quad u(3,0) = 3$

Also $u_t(x,0) = 0 \Rightarrow u_{i,1} = \frac{1}{2}(u_{i-1,0} + u_{i+1,0})$

From this we can find u_{11}, u_{21}, u_{31}

$$u_{11} = \frac{1}{2}(u_{00} + u_{20}) = 2$$

$$u_{21} = \frac{1}{2}(u_{10} + u_{30}) = 3$$

$$u_{31} = \frac{1}{2}(u_{20} + u_{40}) = 2$$

Construct a table by x varies from 0 to 4 with space length of 1

Eqn of the vibrating string of length l is $u_{tt} = c^2 u_{xx}$.

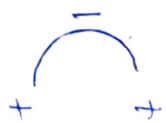
$\therefore$ Its period of vibration $= \frac{2l}{c} = \frac{2(4)}{2} = 4$ sec

required to compute upto $t = 4$ ~~(half period)~~

Half period of vibration is 2 with spacing $k = 1/2$

t \ x		0	1	2	3	4
0	0	$u_{00}=0$	$u_{10}=3$	$u_{20}=4$	$u_{30}=3$	$u_{40}=0$
$\frac{1}{2}$	1	$u_{01}=0$	$u_{11}=2$	$u_{21}=3$	$u_{31}=2$	$u_{41}=0$
1	2	$u_{02}=0$	$u_{12}=0$	$u_{22}=0$	$u_{32}=0$	$u_{42}=0$
$1\frac{1}{2}$	3	$u_{03}=0$	$u_{13}=-2$	$u_{23}=-3$	$u_{33}=-2$	$u_{43}=0$
2	4	$u_{04}=0$	$u_{14}=3$	$u_{24}=-4$	$u_{34}=-3$	$u_{44}=0$

$\frac{1}{2}$ period of vibration.



For III row.

$u_{12} = u_{0,1} + u_{21} - u_{10}$.

$u_{22} = u_{11} + u_{31} - u_{20}$.

$u_{32} = u_{21} + u_{41} - u_{30}$.

Illy for remaing rows

③ Solve the boundary value problem $u_{tt} = u_{xx}$ with the conditions $u(0,t) = u(1,t) = 0$, $u(x,0) = \frac{1}{2}x(1-x)$ & $u_t(x,0) = 0$ taking $h = k = 0.1$ for $0 \le t \le 0.4$

Soln: $c^2 = 1$,

$u(0,t) = u(1,t) = 0 \Rightarrow$ I & last columns are zero.

$u(x,0) = \frac{1}{2}x(1-x) \Rightarrow$

$x = 0.1 \Rightarrow u = 0.045$ $x = 0.5 \Rightarrow u = 0.125$ $x = 0.9 \Rightarrow u = 0.045$

$x = 0.2 \Rightarrow u = 0.08$ $x = 0.6 \Rightarrow u = 0.12$

$x = 0.3 \Rightarrow u = 0.1050$ $x = 0.7 \Rightarrow u = 0.1050$

$x = 0.4 \Rightarrow u = 0.12$ $x = 0.8 \Rightarrow u = 0.08$

t \ x		0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
		0	1	2	3	4	5	6	7	8	9	10
0	0	$u_{00} = 0$	$u_{10} = 0.0450$	$u_{20} = 0.08$	$u_{30} = 0.1050$	$u_{40} = 0.12$	$u_{50} = 0.125$	$u_{60} = 0.12$	$u_{70} = 0.1050$	$u_{80} = 0.08$	$u_{90} = 0.045$	0
0.1	1	u_{01}	$u_{11} = 0.04$	$u_{21} = 0.075$	$u_{31} = 0.1$	$u_{41} = 0.1125$	$u_{51} = 0.12$	$u_{61} = 0.1150$	$u_{71} = 0.1$	$u_{81} = 0.04$	$u_{91} = 0.08$	
0.2	2	u_{02}	$u_{12} = 0.03$	$u_{22} = 0.06$	$u_{32} = 0.0825$	$u_{42} = 0.1$	$u_{52} = 0.1025$	$u_{62} = 0.1$	$u_{72} = 0.05$	$u_{82} = 0.1$	u_{92}	
0.3	3	u_{03}	u_{13}	u_{23}	u_{33}	u_{43}	u_{53}	u_{63}	u_{73}	u_{83}	u_{93}	0
0.4	4	u_{04}	u_{14}	u_{24}	u_{34}	u_{44}	$u_{54} = 0$	u_{64}	u_{74}	u_{84}	u_{94}	0

x / t	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
	0	1	2	3	4	5	6	7	8	9	10
0	0	$u_{00}=0$	0.045 u_{10}	0.08 u_{20}	0.1050 u_{30}	0.12 u_{40}	0.125 u_{50}	0.12 u_{60}	0.1050 u_{70}	0.08 u_{80}	0.045 u_{90}	0 u_{100}
0.1	1	$u_{01}=0$	u_{11} 0.04	0.075 u_{21}	0.1 u_{31}	0.1150 u_{41}	0.12 u_{51}	0.1150 u_{61}	0.1 u_{71}	0.075 u_{81}	0.04 u_{91}	u_{101}
0.2	2	$u_{02}=0$	0.03 u_{12}	0.06 u_{22}	0.085 u_{32}	0.1 u_{42}	0.1050 u_{52}	0.1 u_{62}	0.085 u_{72}	0.06 u_{82}	0.03 u_{92}	0 u_{102}
0.3	3	$u_{03}=0$	0.02 u_{13}	0.04 u_{23}	0.06 u_{33}	0.075 u_{43}	0.08 u_{53}	0.075 u_{63}	0.06 u_{73}	0.04 u_{83}	0.02 u_{93}	0 u_{103}
0.4	4	$u_{04}=0$	u_{14}	u_{24}	u_{34}	u_{44}	u_{54}	u_{64}	u_{74}	u_{84}	u_{94}	0 u_{104}

For I row insert $\frac{1}{2}x(1-x)$

For II row insert $\frac{1}{2}(x+y)$

For III row insert $x+y-M$

} eqns in calculator.

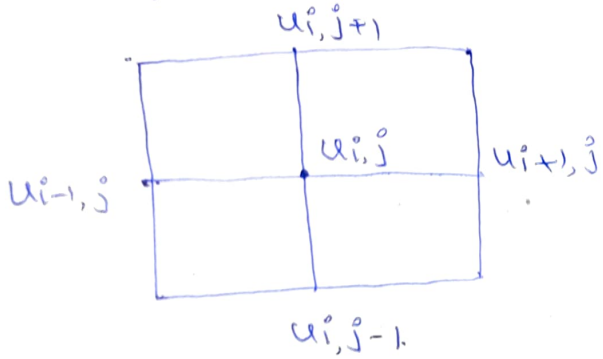
Solution of 2 dimensional Laplace equation (Elliptic eqn)

Suppose x & y are 2 independent variables & u is a real valued function of x & y. Then PDE $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is known as 2 dimensional Laplace eqn.

Note: To solve this (1) we use standard 5-point formula (S.F) i.e.

$$u_{i,j} = \frac{1}{4}\left[u_{i-1,j} + u_{i+1,j} + u_{i,j+1} + u_{i,j-1}\right] \longrightarrow (1)$$

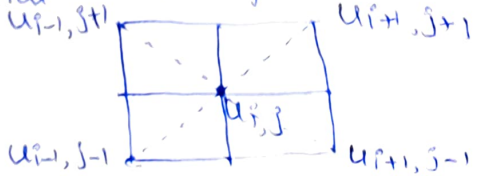
i.e $u_{i,j}$ at any interior mesh point is the average of its values at 4 neighbouring points to the left, right, above & below.



(2) We can use diagonal 5 point formula (D.F)

$$u_{i,j} = \frac{1}{4}\left[u_{i-1,j+1} + u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j-1}\right] \longrightarrow (2)$$

i.e $u_{i,j}$ is the average of the values at 4 neighbouring diagonal mesh points.



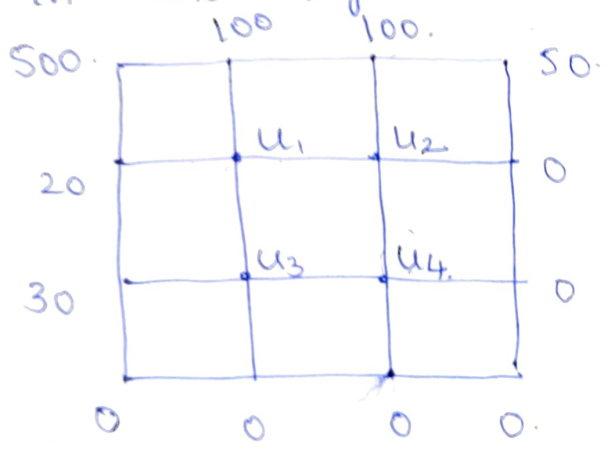
Sf is more accurate than D.F

(3) The accuracy of the value can be improved by successive application of the formula (3) we regard the obtained values of $u_{i,j}$ as the initial approximation denoted by $u_{i,j}^{(0)}$, then the first iteration is carried out by using the latest iterative value available on hand, where we have i.e

$$u_{i,j}^{(1)} = \frac{1}{4}\left[u_{i-1,j}^{(0)} + u_{i+1,j}^{(0)} + u_{i,j-1}^{(0)} + u_{i,j+1}^{(0)}\right]$$

It is repeated till we get the degree of accuracy. This process is called Liebmann's iteration process or Gauss-Seidel process.

① Solve $u_{xx} + u_{yy} = 0$ in the following square region with the boundary conditions as in the fig.



Apply standard 5 point formula

Let $u_4 = 0$.

$u_1 = \frac{1}{4}[500 + 30 + 100 + 0] \Rightarrow \boxed{u_1 = 157.5}$ [DF]

$u_2 = \frac{1}{4}[\overset{u_1}{157.5} + 100 + 0 + \overset{u_4}{0}] \Rightarrow \boxed{u_2 = 64.25}$ (SF)

$u_3 = \frac{1}{4}[30 + \overset{u_4}{0} + \overset{u_1}{157} + 0] \Rightarrow \boxed{u_3 = 46.75}$ (SF)

$\boxed{u_4 = 0}$ $\frac{1}{4}[u_3 + u_2 + 0 + 0] = \frac{1}{4}[46.75 + 64.25 + 0]$ [SF] $= 27.75$

Now $u_1^{(0)} = 157.5$, $u_2^{(0)} = 64.25$, $u_3^{(0)} = 46.75$

$u_4^{(0)} = 27.75$

Apply ~~diagonal~~ SF formula

$u_1 = \frac{1}{4}[20 + 100 + u_2 + u_3]$

$\Rightarrow 4u_1 - u_2 - u_3 = 120$.

$A = \frac{1}{4}(120 + B + C)$.

$u_2 = \frac{1}{4}[u_1 + 0 + 100 + u_4]$

$\Rightarrow 4u_2 - u_1 - u_4 = 100$.

$B = \frac{1}{4}(100 + A + D)$

$u_3 = \frac{1}{4}[30 + u_1 + u_4 + 0]$

$4u_3 - u_1 - u_4 = 30$.

$C = \frac{1}{4}(30 + u_1 + u_4)$

$A + D$

$u_4 = \frac{1}{4}(u_3 + u_2 + 0 + 0)$

$D = \frac{1}{4}(B + C)$

Insert eqns: $A = \frac{1}{4}(120 + B + C)$; $B = \frac{1}{4}(100 + A + D)$; $C = \frac{1}{4}(30 + A + D)$

$D = \frac{1}{4}(B + C)$

Insert

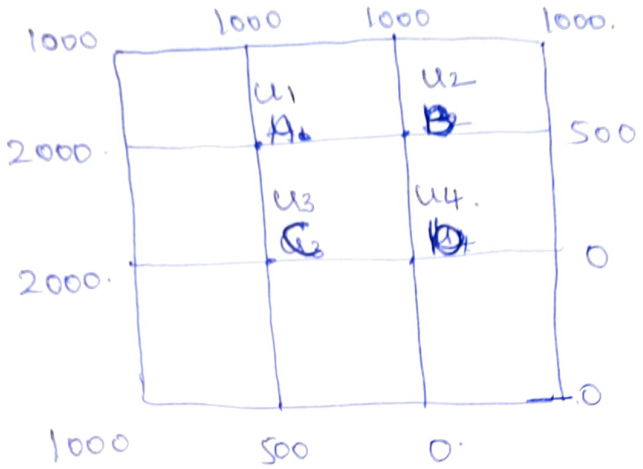
u_1	u_2	u_3	u_4
57.75	46.375	28.875	18.8125
48.8125	41.9063	24.4063	16.5781
46.5781	40.7891	23.2891	16.0195
46.0195	40.5098	23.0098	15.8799
45.8799	40.4399	22.9399	15.8450
45.845	40.4225	22.9225	15.8362
45.8362	40.4181	22.9181	15.8341
45.8341	40.4170	22.9170	15.8335
45.8335	40.4168	22.9168	15.8334
45.8334	40.4167	22.9167	15.8333

Since 9th & 10th iterations are exactly equal upto 3rd decimal places So stop

$\therefore$ $u_1 = 45.8334$ $u_2 = 40.4167$ $u_3 = 22.9167$

$u_4 = 15.8333$.

(2) Given the values of $u(x,y)$ on the boundary of the square in the fig given below, evaluate the fun $u(x,y)$ satisfying the Laplace eqn $\nabla^2 u = 0$ at the pivotal points of this fig.



To get the initial values of A, B, C & D. we assume u_4. $D = 0$. Then.

$$A = \frac{1}{4}(2000 + 1000 + 1000 + 0) \quad (DF)$$

$$\Rightarrow \boxed{A^{(0)} = 1000}$$

$$B = \frac{1}{4}(A + D + 1000 + 500)$$

$$B = \frac{1}{4}(1000 + 0 + 1000 + 500) \quad (SF)$$

$$\boxed{B^{(0)} = 625}$$

$$C = \frac{1}{4}(2000 + 500 + A + D) \quad (SF)$$

$$= \frac{1}{4}(2000 + 500 + 1000 + 0)$$

$$\boxed{C^{(0)} = 875}$$

$$D = \frac{1}{4}(C + B + 0 + 0)$$

$$= \frac{1}{4}(875 + 625 + 0) =$$

$$D = 375$$

$$\therefore \quad A^{(0)} = 1000, \quad B^{(0)} = 625, \quad C^{(0)} = 875, \quad D^{(0)} = 375$$

Now

$$A^{(1)} = \frac{1}{4}(2000 + 1000 + C + B)$$

$$A = 4^{-1}(3000 + B + C)$$

$$B^{(1)} = \frac{1}{4}(1000 + 500 + A + D)$$

$$B^{(1)} = 4^{-1}(1500 + A + D)$$

$$C^{(1)} = \frac{1}{4}(2000 + A + D + 500)$$

$$C^{(1)} = 4^{-1}(2500 + A + D)$$

$$D^{(1)} = \frac{1}{4}(B + C + 0)$$

$$\Rightarrow D = 4^{-1}(B + C).$$

Insert all the eqns in calculator.

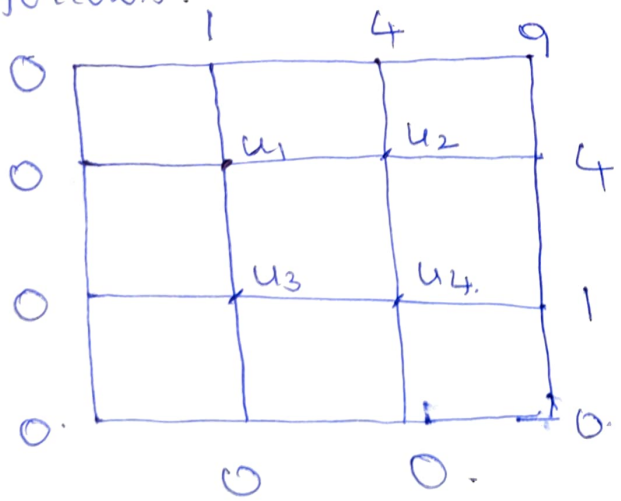
$A = 4^{-1}(3000 + B + C)$: $B = 4^{-1}(1500 + A + D)$: $C = 4^{-1}(2500 + A + D)$

$D = 4^{-1}(B + C)$

A	B	C	D
1125	750	1000	437.5
1187.5	781.25	1031.25	453.125
1203	789.0625	1039.0625	457.0313
1207.0313	791.0156	1041.0156	458.0078
1208.0078	791.5039	1041.5039	458.25
1208.252	791.62	1041.626	458.313
1208.3130	791.6565	1041.6565	458.3282

HW

③ Solve $\nabla^2 u = 0$ in the following square region with the boundary conditions as follows.



Let $u_4 = 0$.

$u_1 = \frac{1}{4}(0 + 4 + \tilde{u_4} + 0) \Rightarrow \boxed{u_1 = 1}$ (SF)